

II BE EXAMINATION DEC 2023
Information Technology
3ITRC3 Data Structures

Maximum Marks [60]

Time-3 hours

Note-Attempt all parts from Question 1. For rest of the questions, attempt any two parts from each.

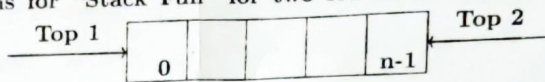
Q.1 (a) Heap is a linear data structure. Yes/No. Support your answer. [1]

(b) You need to take input for your program from a file. How would you read a line from a file in C++. [1]

(c) Give a scenario where Bubble Sort would perform worst. [1]

(d) What is a Stable Sorting algorithm? [1]

(e) What would be conditions for "Stack Full" for two stacks when you implement two stacks using an array? [1]



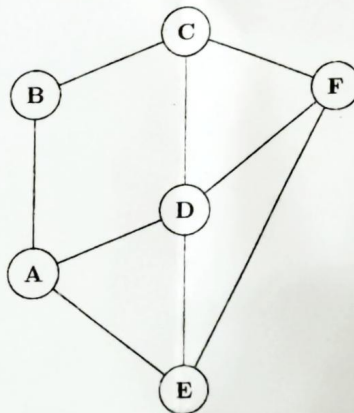
(f) Give a scenario where adjacency list would be a better choice than adjacency graph for representing a graph. [1]

(g) Define a Binary tree. [1]

(h) Suggest two applications of a priority queue. [1]

(i) The maximum number of nodes in a binary tree of depth k is _____. [1]

(j) Write the output when you apply BFS on the graph starting from vertex D. [1]



(k) Why did collision occurs in a Hash Table? [1]

(l) How would you identify number of elements in a circular queue in terms of rear and front. [1]

Q.2 (a) How would you represent a polynomial using an array. Write the procedure/algorithm for adding two polynomials using arrays. [6]

(b) How would you represent a sparse matrix efficiently using an array. Implement a transpose function as per your representation. [6]

(c) Write a recursive procedure copy() to make a copy of a linked list. [6]